DevOps INSTITUTE

ADVANCING THE HUMANS OF DEVOPS

presents

Global SKILup Day

SKILup Session

BMK Lakshminarayanan

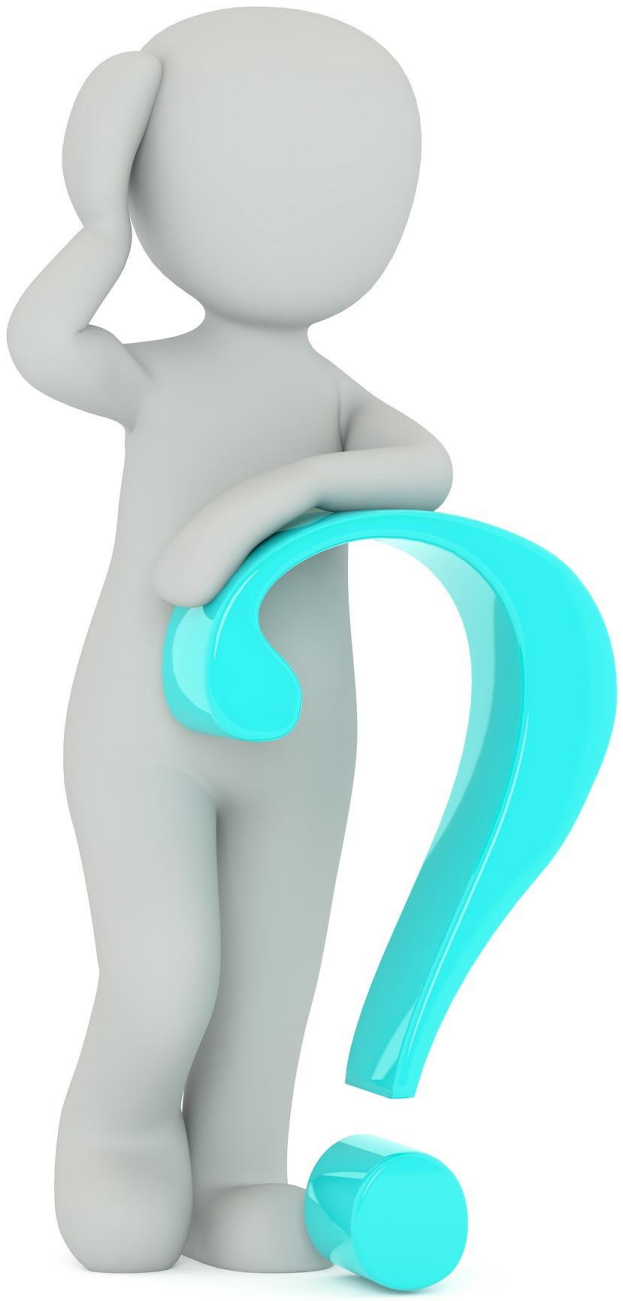
Solutions Architect

Bank of New Zealand

DECEMBER 10, 2019

18-HOUR VIRTUAL EVENT

FREE • ONLINE • DEVOPSINSTITUTE.COM/GLOBALSKILUPDAY



How to Avoid Cloud Confusion, DevOps Dilemma, Microservice Madness

BMK Lakshminarayanan

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ABOUT ME

BMK Lakshminarayanan

Solutions Architect | DevOps Advocate

MISSION

Make your life better,

Make others life better,

Make the world better

Community

DEVOPS DAYS

CLOUD NATIVE SUMMIT

CLOUD NATIVE COMPUTING FOUNDATION AMBASSADOR

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Cloud

DevOps

Microservices

Governance

Cloud-Native

Canary Releases

Kubernetes

Pipeline

NoSQL

Statefulsets

Continuous Delivery

Event-Sourcing

Cloud

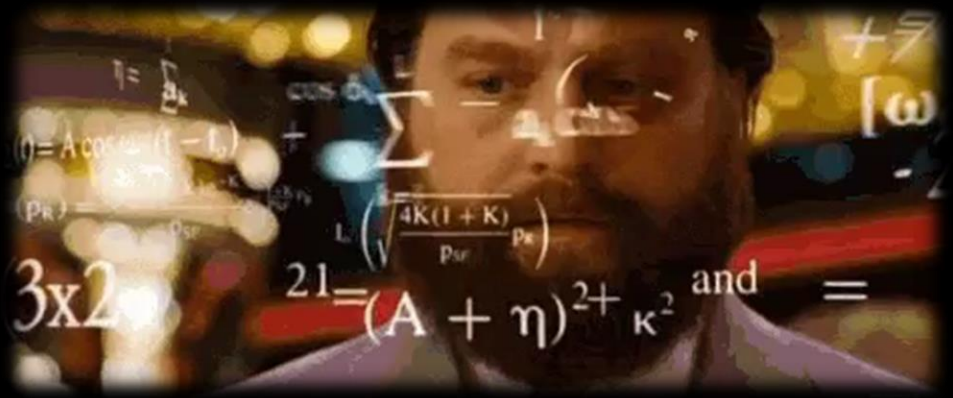
Hybrid-Cloud

Multi-Cloud

Containers

Canary Releases

yaml



Continuous Integration

Microservices

Lean

Observability

TDD

Open Source

DevOps

Scale

DevSecOps

Monoliths

DSC

A/B Testing

CQRS

Continuous Deployments

Agile

Operators

Dark Launch

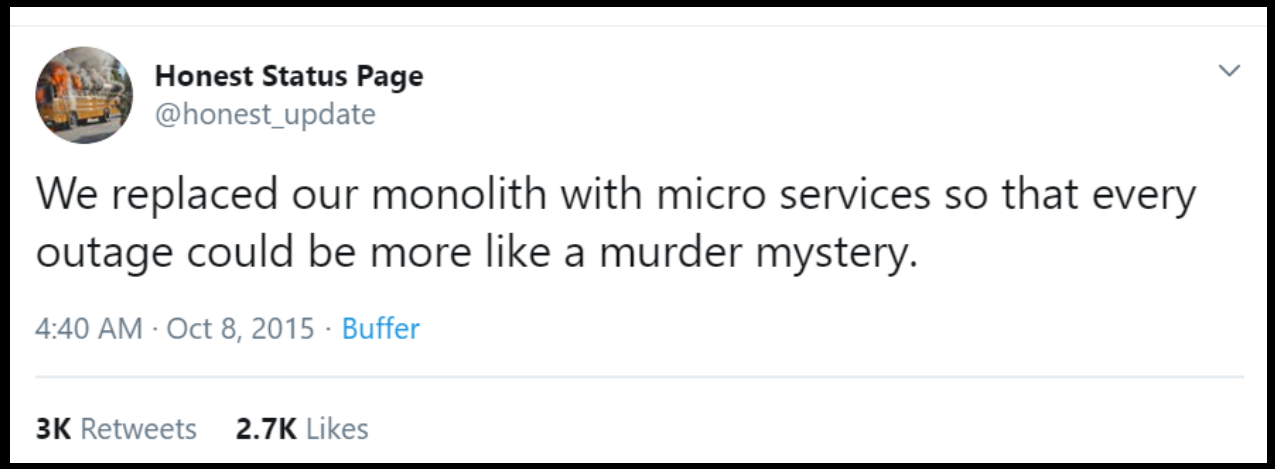
API

Feature Toggles

PaaS

VSM

Microservice Challenges



Microservice Challenges

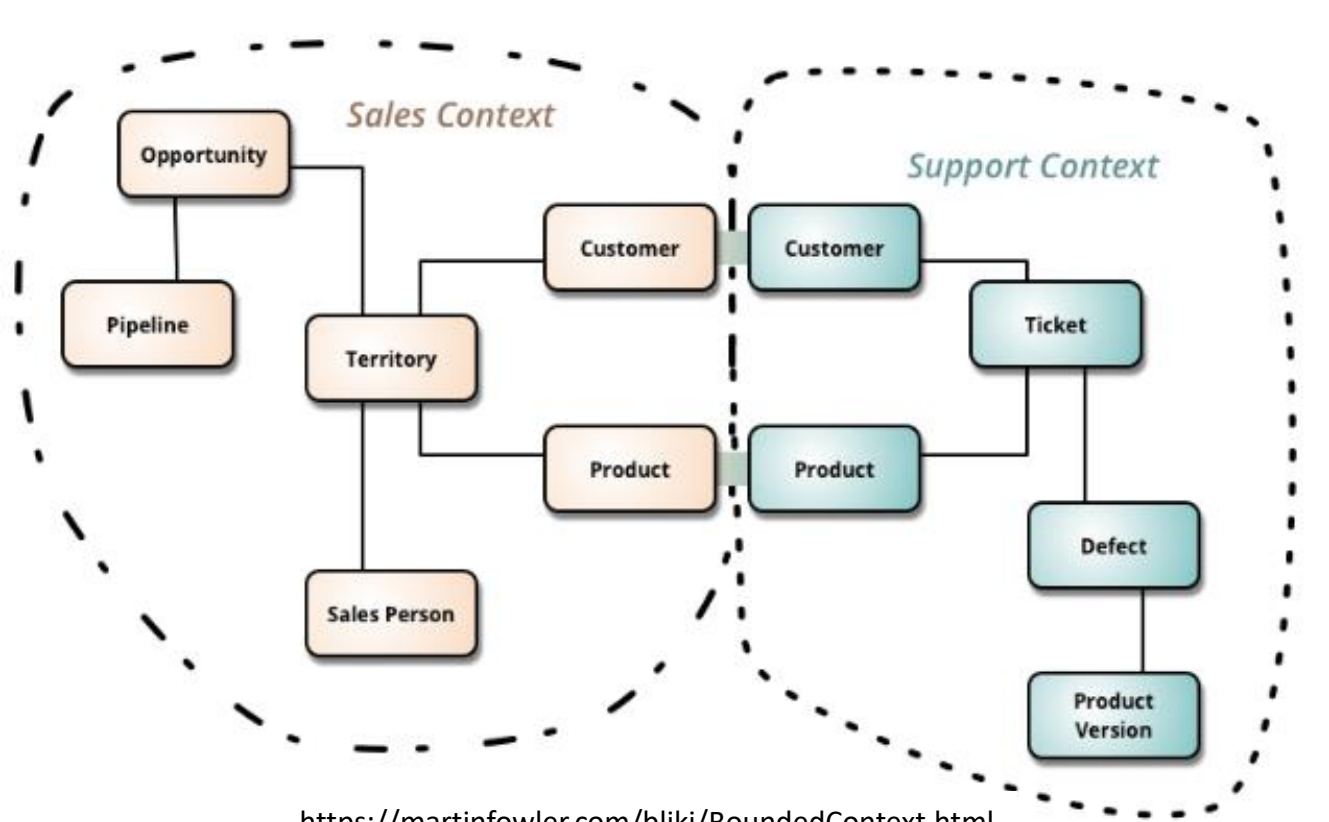
How big, how small? Where to draw the boundary?

Data in Microservice world

Service Communication

How big, how small? Where to draw the boundary?

- How:
  - Apply Domain Driven Design
  - Work with “Domain” experts
  - Understand the Domain, Context, Boundary, Interactions



https://martinfowler.com/bliki/BoundedContext.html

Data in Microservices World?

- How
  - Decentralized Governance
  - Decentralized Data Management
  - Database or Schema / Service Boundary

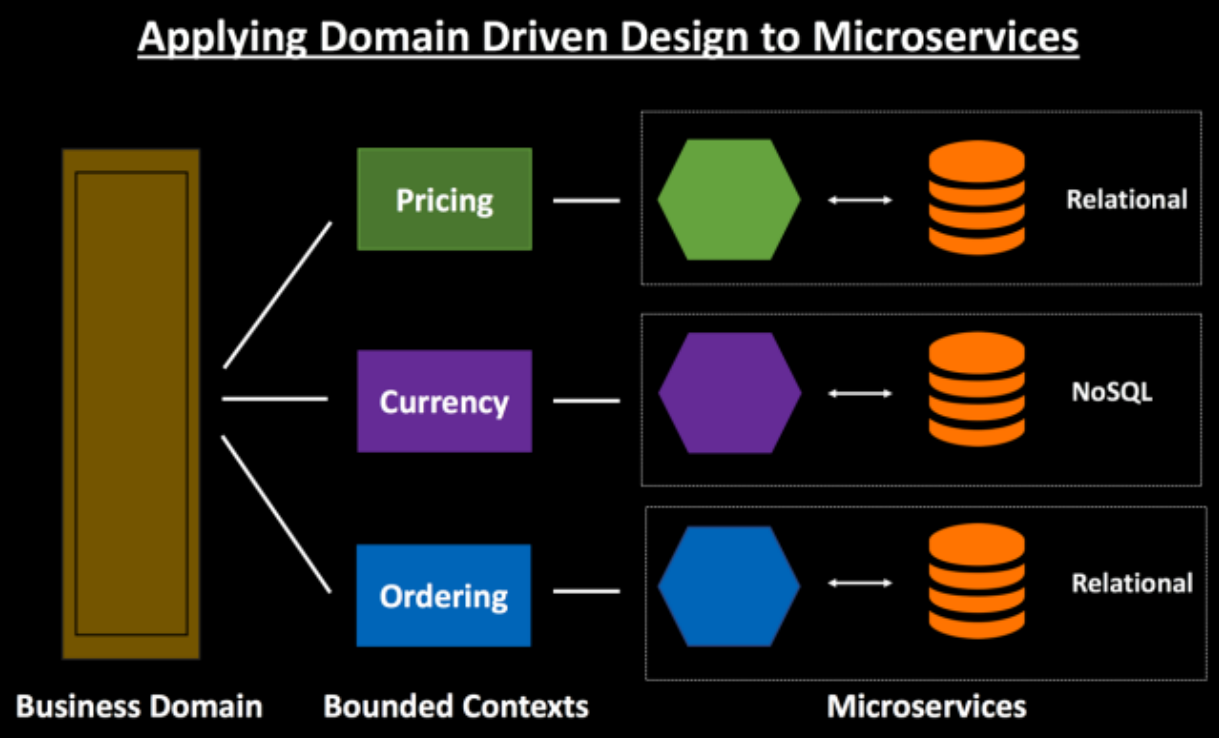
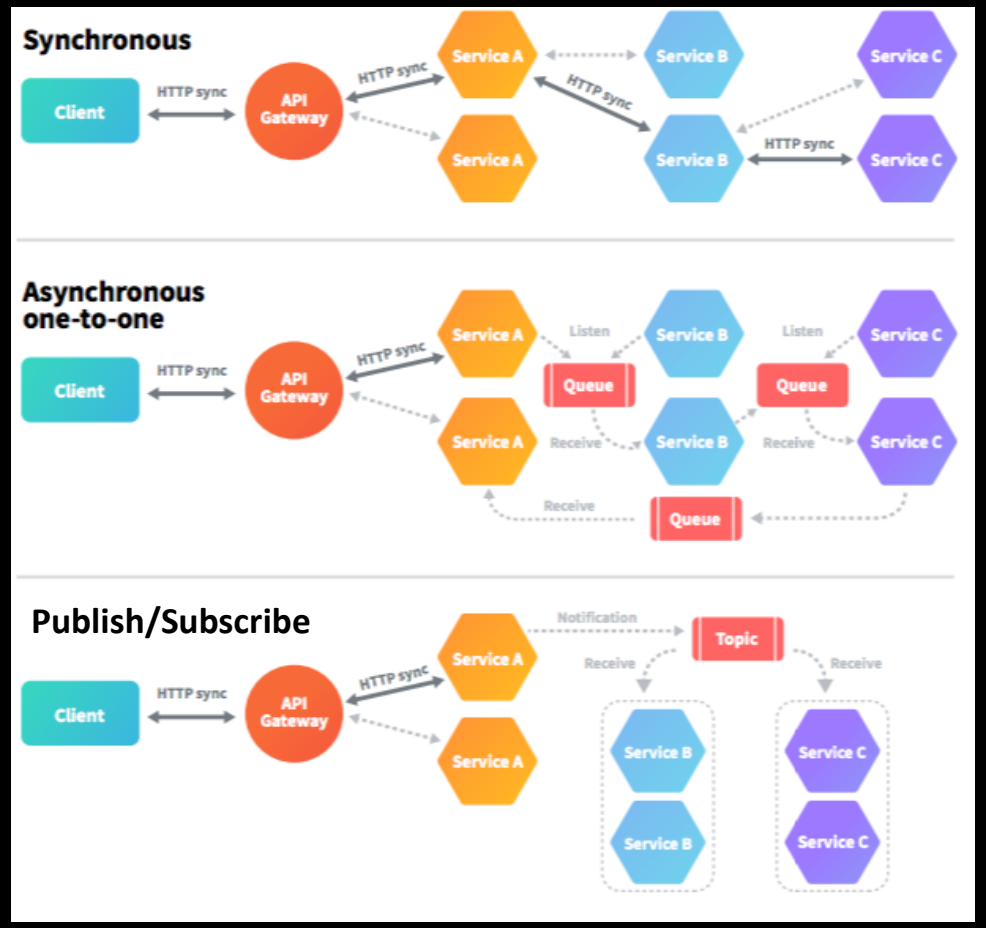


Image Courtesy: https://dzone.com/articles/breaking-the-monolithic-database-in-your-microserv

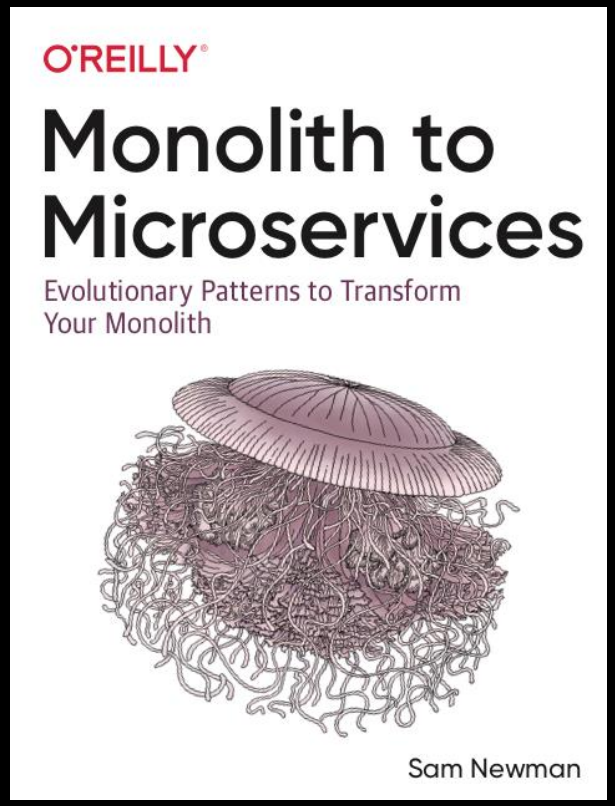
Service-Service Communication

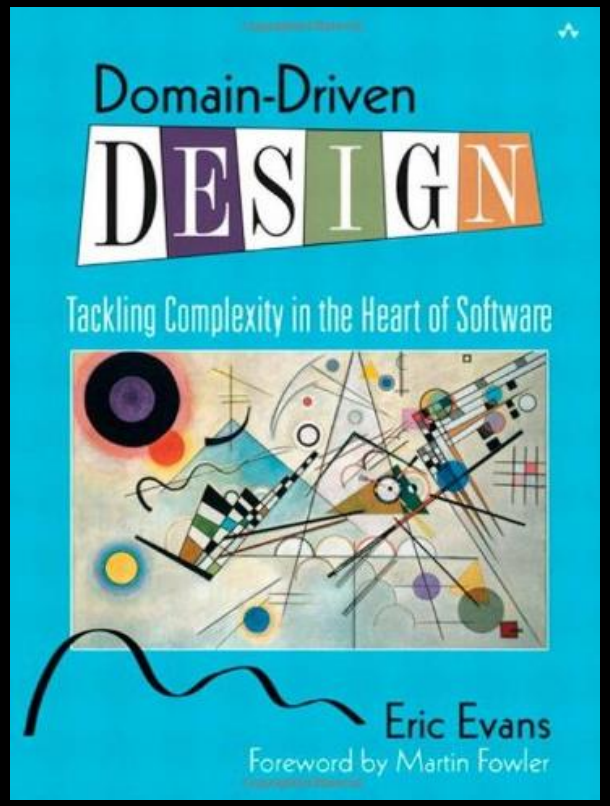
- How
  - Event Driven Architecture
  - Business Events e.g. Like Credit Decision, Loan Approved
  - Platforms: Kafka, NATS

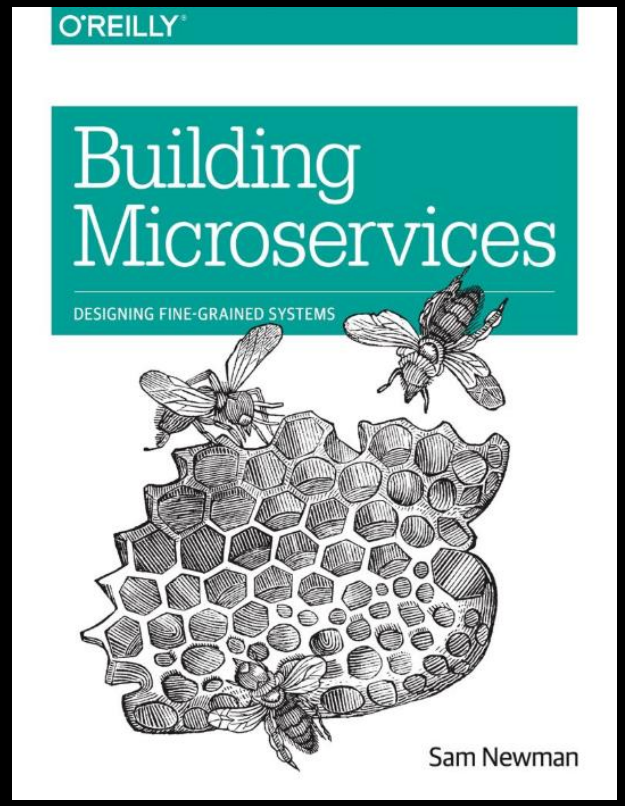


Traps To Avoid

- Everything into Microservices
- Patterns Overloaded (Event-Sourcing, CQRS)
- Too much focus on “Reusability”







Helpful Resources

Cloud Challenges

Cloud Challenges

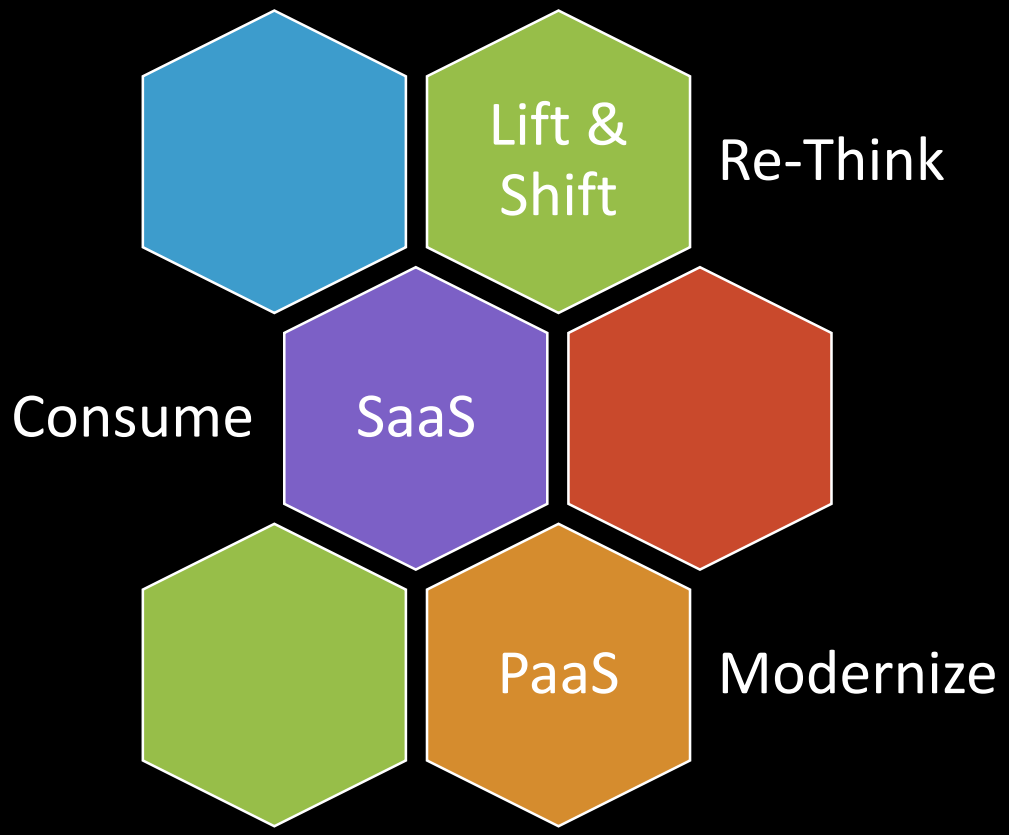
Modernize or not to modernize?

Cloud-Native Architecture

Cloud Platform Team

Modernize?

- How
  - Application assessment – Data, Security, Current state
  - Define Target state: Platform, Services, Tools, Monitoring
  - Refactor to take “Cloud” advantage



Cloud-Native Architecture

- How
  - Automation: Infrastructure provisioning, Application/Service CI/CD, Monitoring health, Recovery
  - Secure: Every component is designed to protect itself
  - Data: Identify, Classify, Design

CNCF defines cloud-native as "scalable applications" running in "modern dynamic environments" that use technologies such as containers, microservices, and declarative APIs.

CLOUD NATIVE COMPUTING FOUNDATION

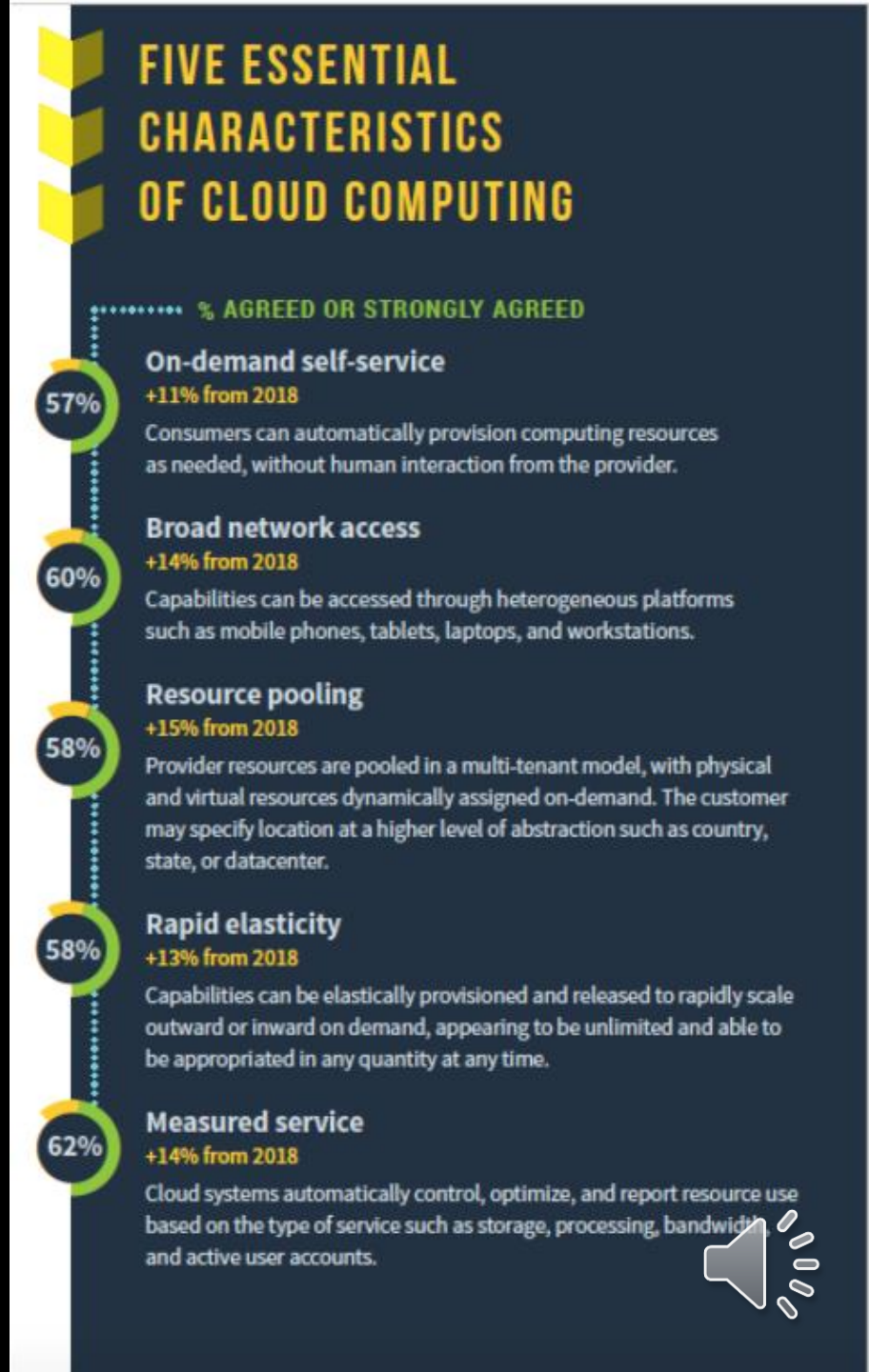
https://www.cncf.io/

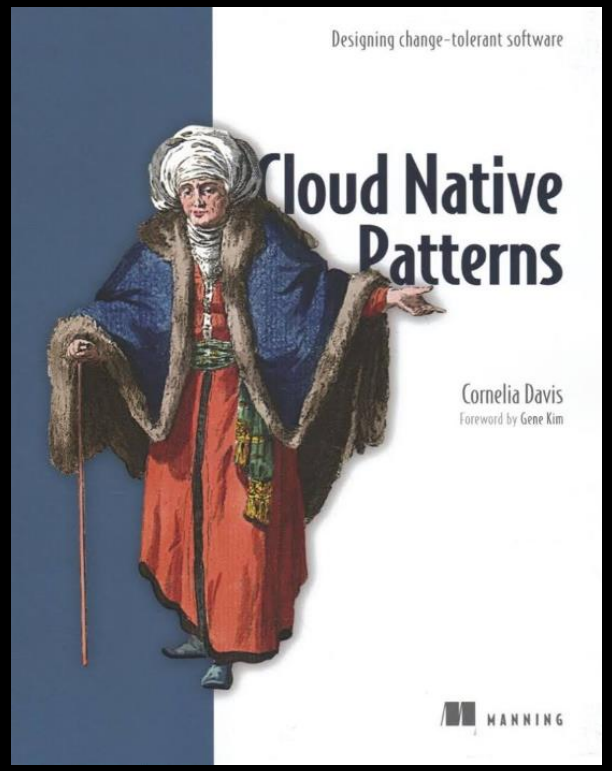
Cloud Platform Team

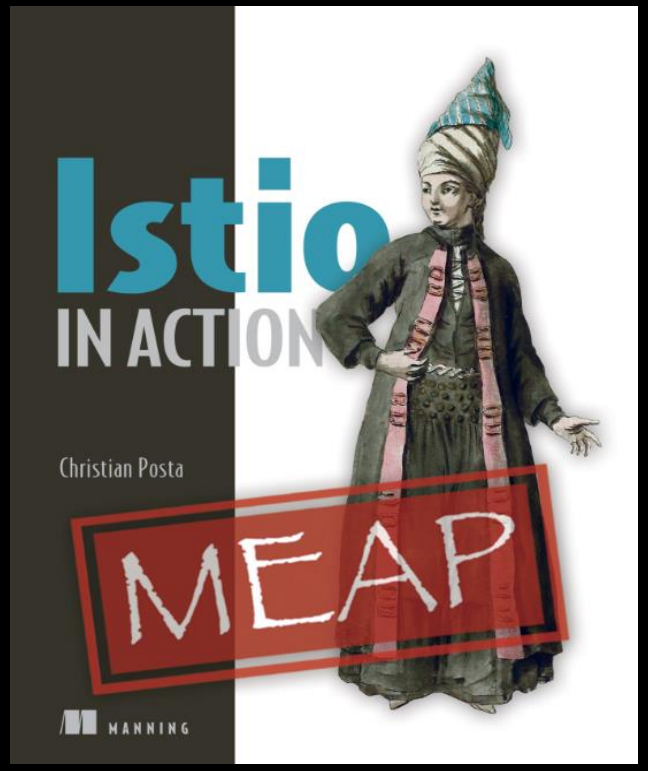
- How
  - Platform Team – Similar to “Product” teams
  - Enable “Self-Service”
  - Advocates and Evangelist

Traps To Avoid

- Tickets to create manual “Cloud” resources
- “Governance” – Too much Control & Command
- Overloading the team







Helpful Resources

DevOps Challenges

DevOps Challenges

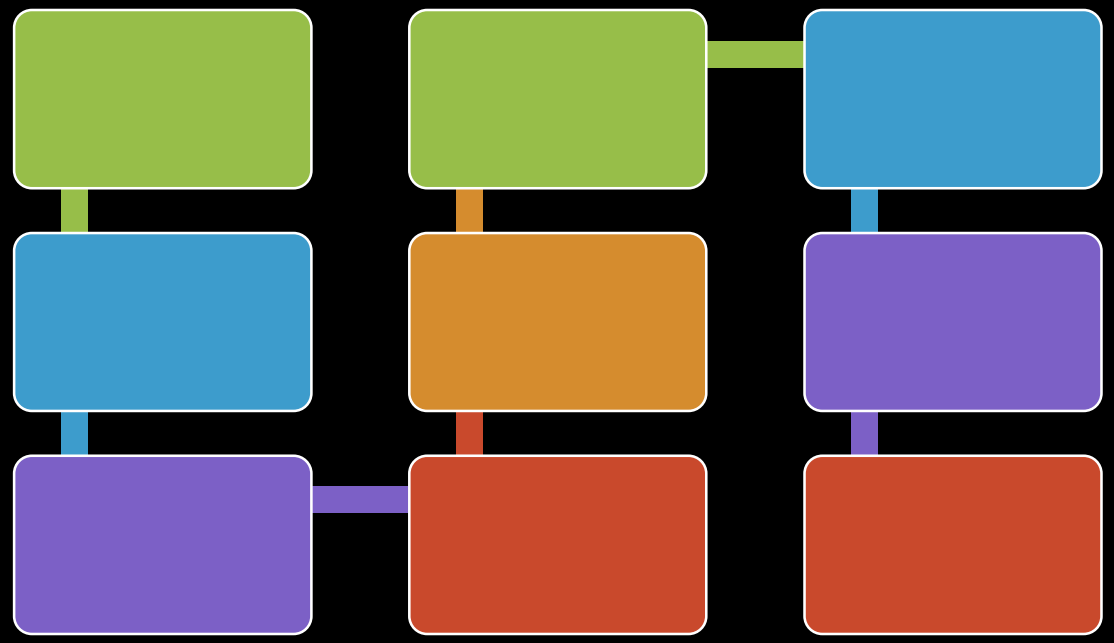
Tools focused & Disconnected Toolchain

Pipeline focused

Centralized “DevOps” Team

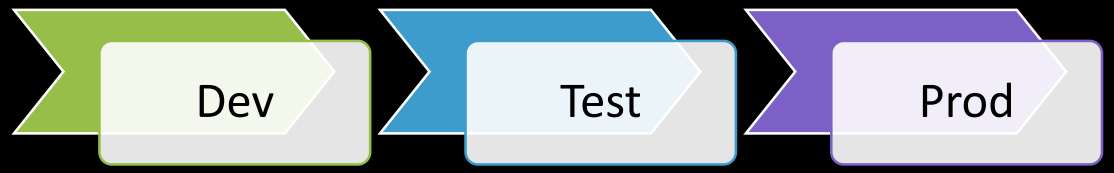
Tools Focused, Disconnected Toolchain

- How
  - Value Stream Mapping
  - Establish Connected Toolchain
  - Review regularly

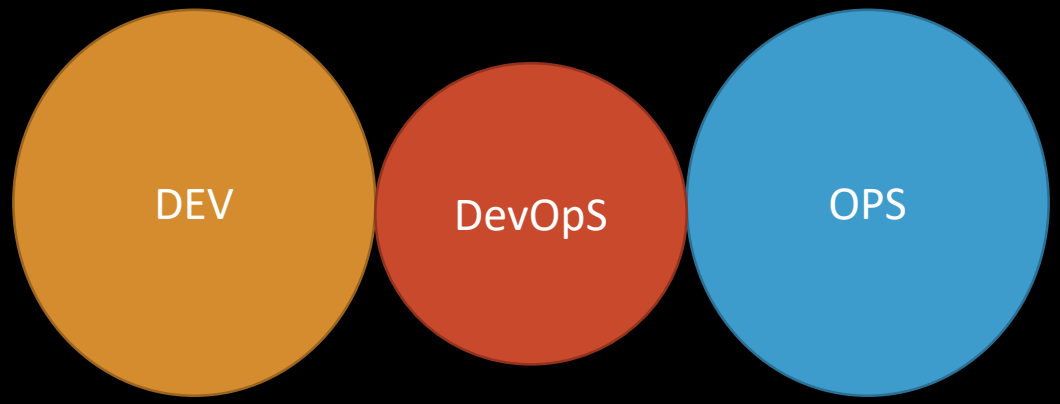


Pipeline Focused

- How
  - Establish “Guidance” and “Guidelines”
  - Set expectations as “Gates”
  - Aim for “Fully” automated pipelines



Centralized “DevOps” Team: Apply Team Design Principles



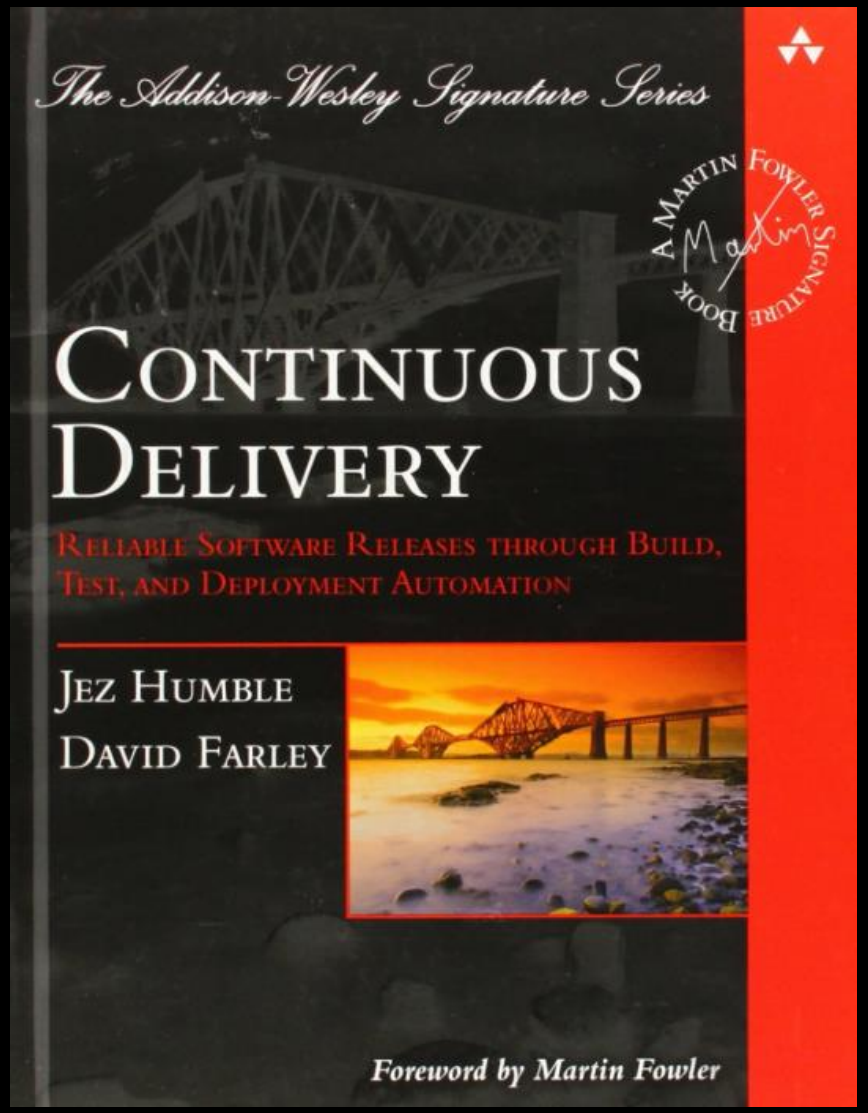
The DevOps Team Silo (Anti-Type B) typically results from a manager or exec deciding that they “need a bit of this DevOps thing” and starting a ‘DevOps team’ (probably full of people known as ‘a DevOp’).

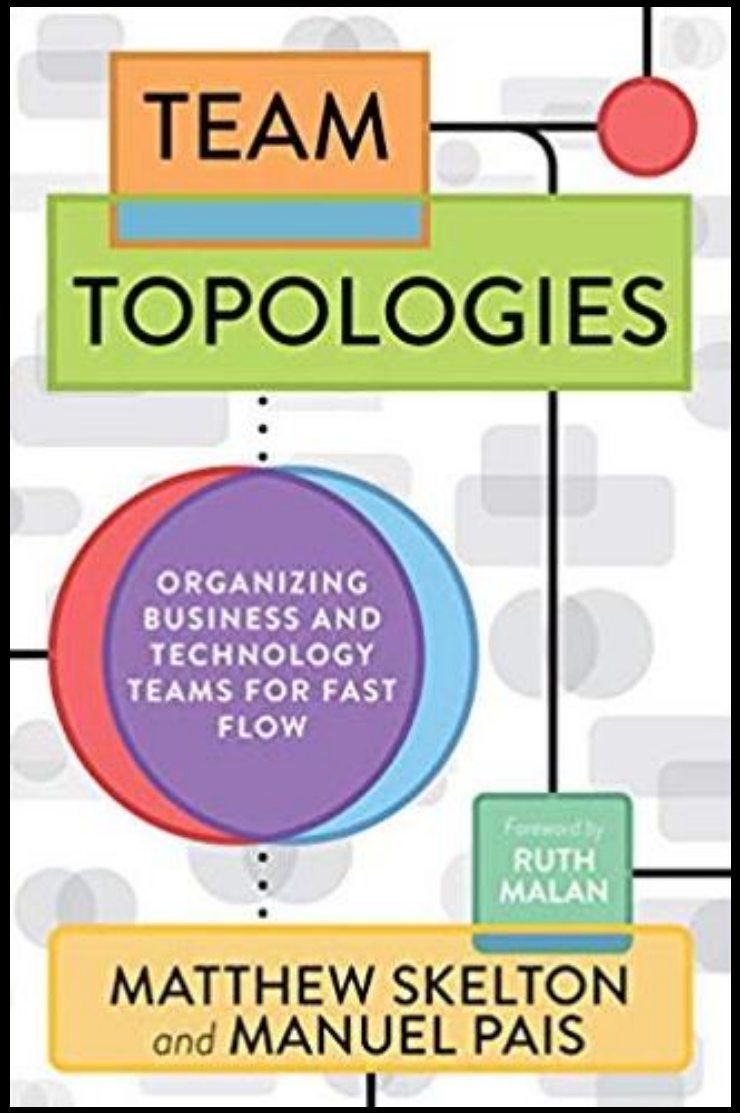
The members of the DevOps team quickly form another silo, keeping Dev and Ops further apart than ever as they defend their corner, skills, and toolset from the ‘clueless Devs’ and ‘dinosaur Ops’ people.

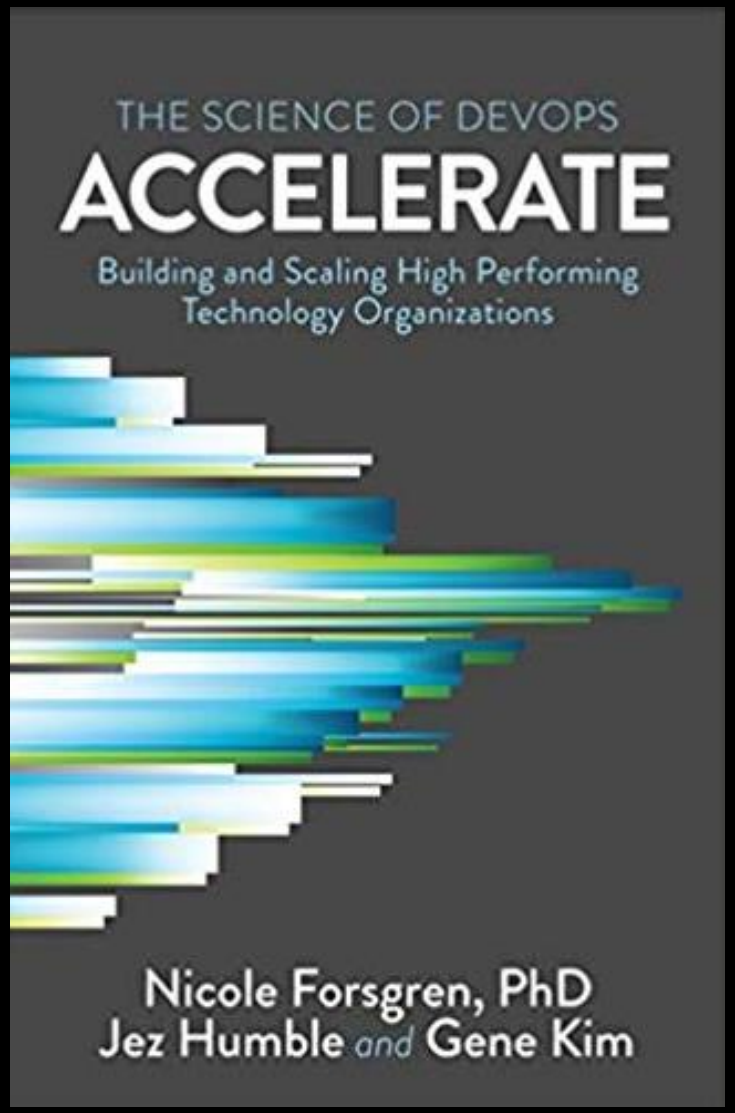
– *Team Topologies*

Traps To Avoid

- DevOps Project
- DevOps Team
- Autonomy Misunderstanding

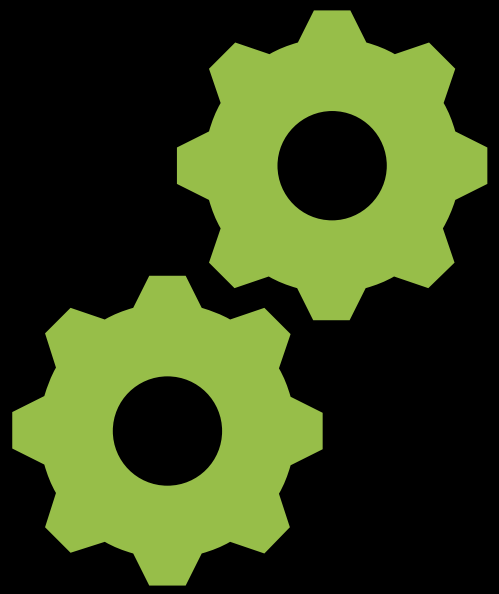






Helpful Resources

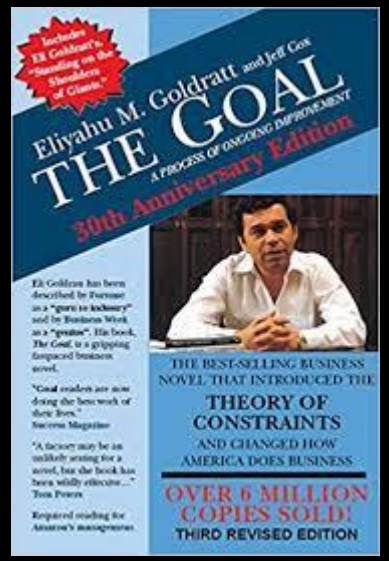
DEVOPS IS NOT A GOAL, BUT A NEVER-ENDING PROCESS OF CONTINUAL IMPROVEMENT
- Jez Humble



Culture
Automation
Lean
Metrics
Share

Some More Ideas

- Ubiquitous language
- Disciplined & Focused Team
- Sensible Architecture
- Upskill
- Reduce Context Switch
- Developer Productivity


Eliyahu M. Goldratt and Jeff Cox
THE GOAL
A PROCESS OF ONGOING IMPROVEMENT
30th Anniversary Edition
THE BEST-SELLING BUSINESS NOVEL THAT INTRODUCED THE
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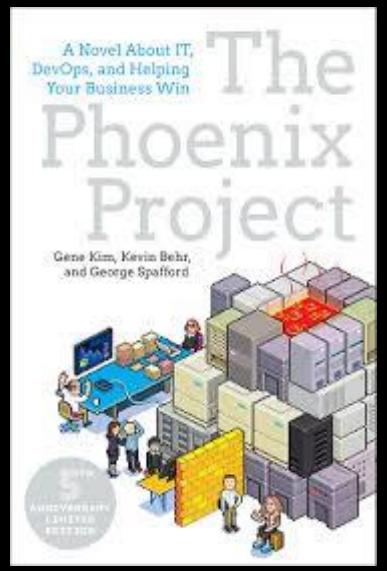
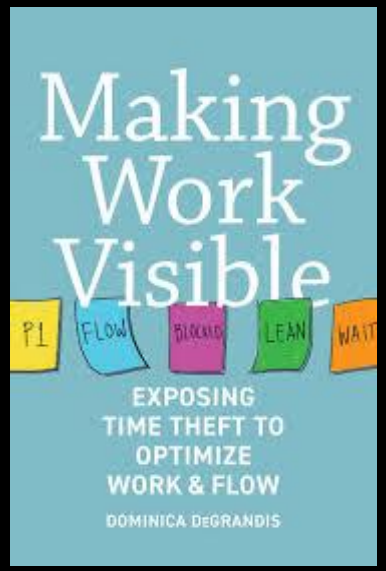
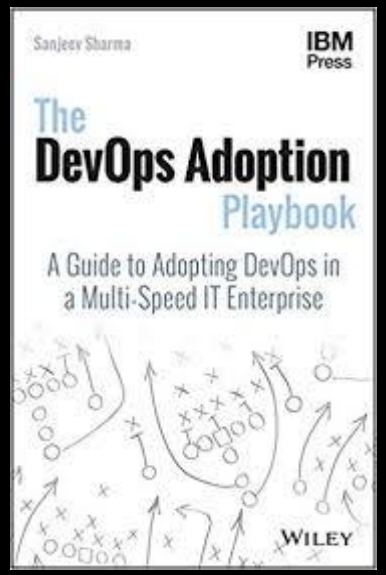
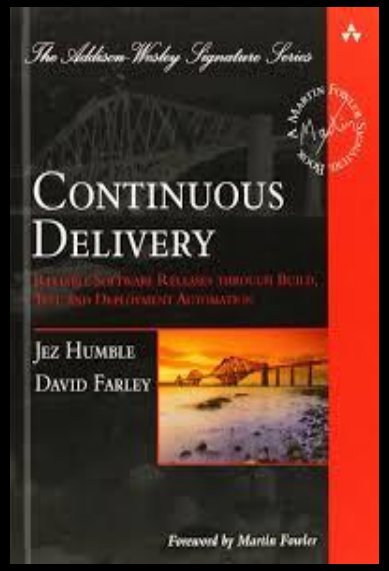
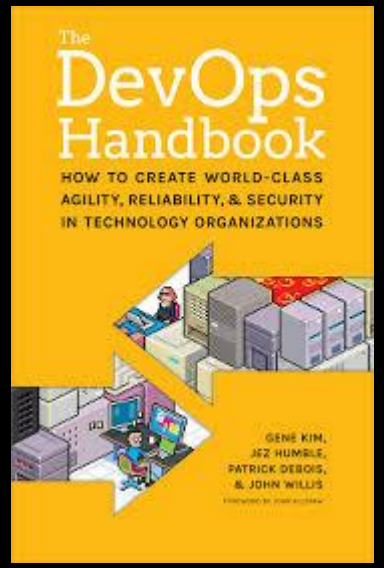

A Novel About IT, DevOps, and Helping Your Business Win
The Phoenix Project
Gene Kim, Kevin Behr, and George Spafford

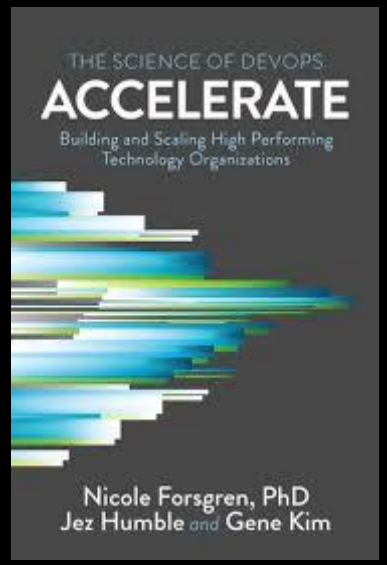

Making Work Visible
PI
FLOW
BLOCKED
LEAN
WAIT
EXPOSING TIME THEFT TO OPTIMIZE WORK & FLOW
DOMINICA DeGRANDIS

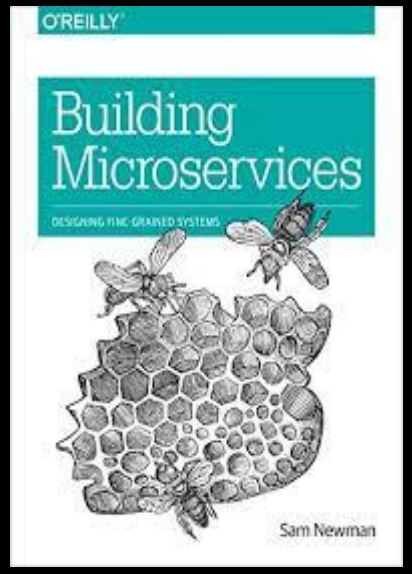

Sanjeev Sharma
IBM Press
The DevOps Adoption Playbook
A Guide to Adopting DevOps in a Multi-Speed IT Enterprise
WILEY

PROJECT TO PRODUCT
MIK KERSTEN
FOREWORD BY GENE KIM


The Addison-Wesley Signature Series
CONTINUOUS DELIVERY
JEZ HUMBLE
DAVID FARLEY
Foreword by Martin Fowler


The DevOps Handbook
HOW TO CREATE WORLD-CLASS AGILITY, RELIABILITY, & SECURITY IN TECHNOLOGY ORGANIZATIONS
GENE KIM, JEZ HUMBLE, PATRICK DEBOIS, & JOHN WILLIS


THE SCIENCE OF DEVOPS
ACCELERATE
Building and Scaling High Performing Technology Organizations
Nicole Forsgren, PhD
Jez Humble and Gene Kim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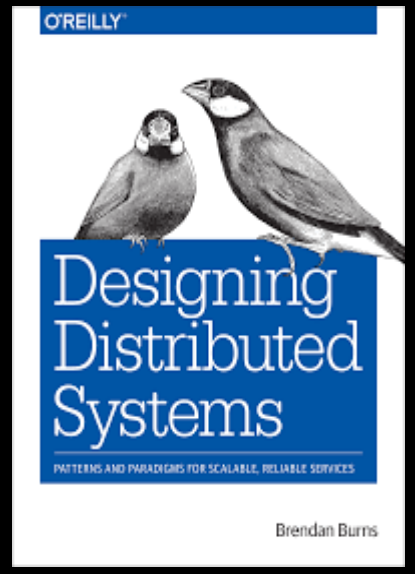
DESIGNING FINE-GRAINED SYSTEMS
Sam Newman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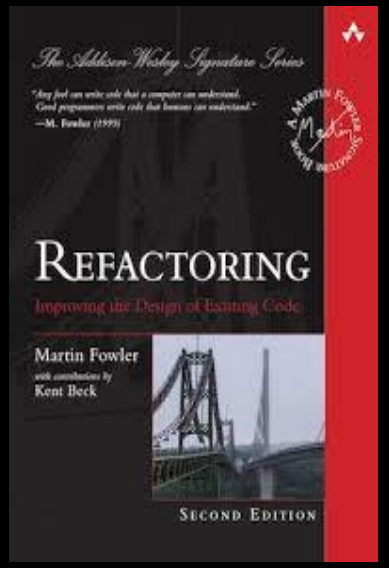
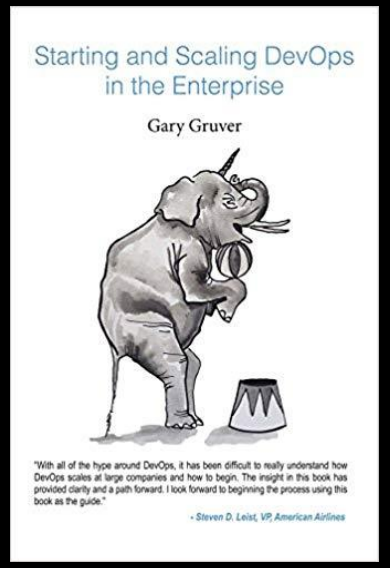
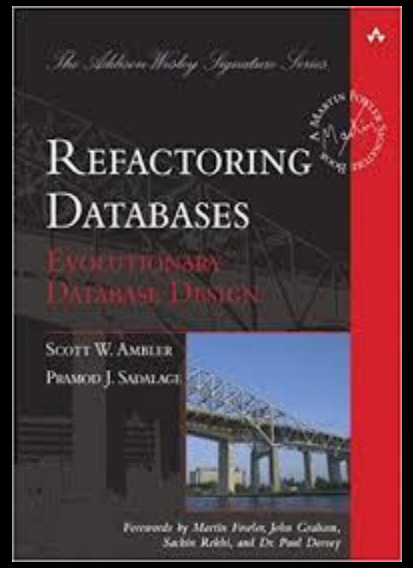
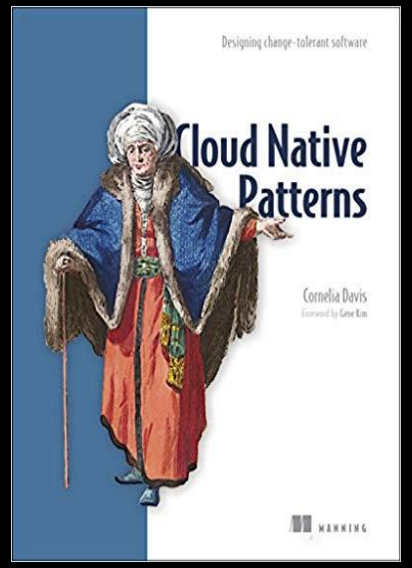
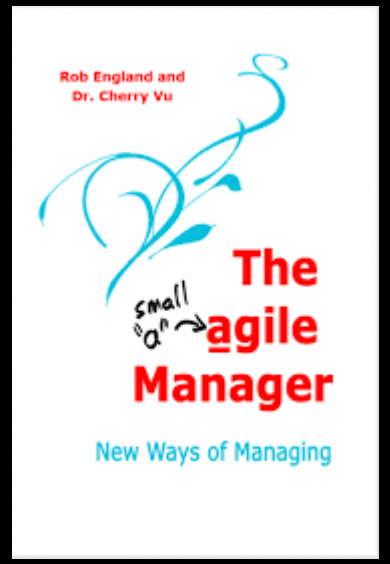
Designing Distributed Systems
PATTERNS AND PARADIGMS FOR SCALABLE, RELIABLE SERVICES
Brendan Burns











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